Home Credit Default Risk Prediction Tool

An ML-Based Solution for Smarter Lending Decisions



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Overview of the Presentation

- **Business Problem & Objective:** Credit default risk and the need for better risk assessment.
- **Data & Quality Insights:** Key patterns identified from customer data.
- **Machine Learning Solution:** Model development, comparison, and evaluation.
- **Risk Prediction :** Defaulters and Flag for Review decisions.
- **Potential Enhancements :** Improving risk assessment with additional information and insights.

Business Problem & Objective

★ Home Credit is a consumer finance company focused on financial inclusion, helping underserved customers—including people with little or no credit history—access financing responsibly.

★ The challenge: How can we better assess an applicant's repayment risk when traditional credit information is limited?

★ Our goal: a model that tells Home Credit, at the time of application, how likely a client is to repay — so mistakes happen less often.

Reject a good client

Many Home Credit applicants have little or no formal credit history. Turn away too many of them, and the business loses interest income and market share to competitors willing to take a chance.

Lost revenue

Approve a risky client

Approve someone who can't keep up with payments, and Home Credit absorbs collections costs and write-offs on a loan that should never have been issued.

Direct financial loss

The Task and Who Acts On It

✓ **CLASSIFICATION (our approach)**

Will this client repay, or struggle to repay?

We predict a probability of payment difficulty (0-1) for every application — the number underwriters act on.

✗ **REGRESSION (not this)**

Predicting a number (e.g. loss amount)

We're deciding whether to approve at all — a category to act on, not a dollar figure to forecast.

WHO ACTS ON THIS

Home Credit's Underwriting & Credit Risk Team

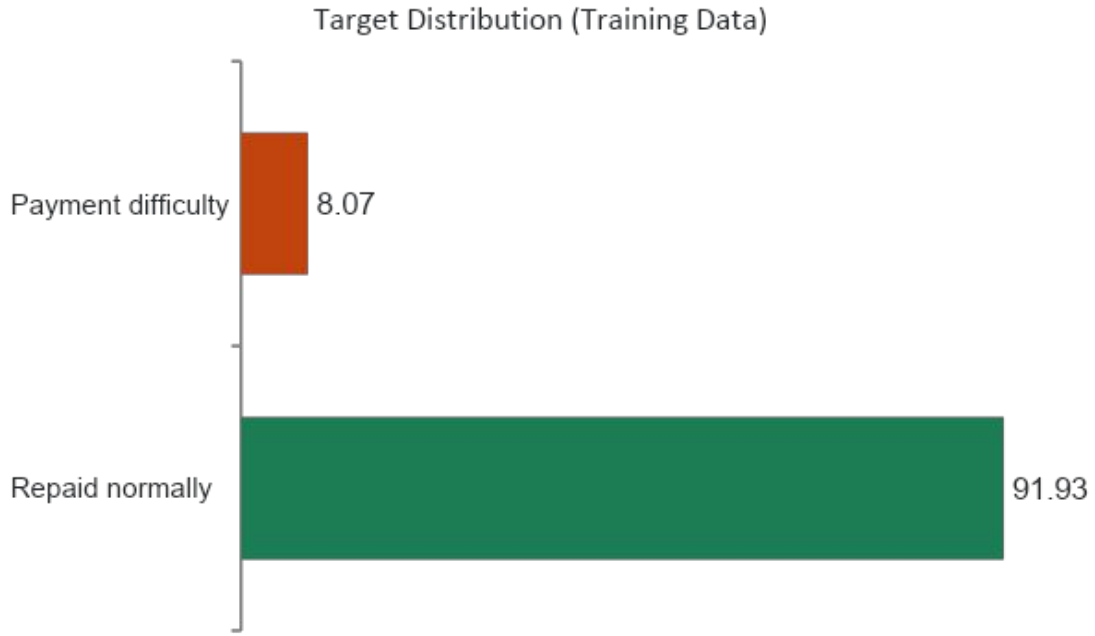
Reviews every application, decides approve / decline / refer, and sets the loan amount and repayment period — measured on both default rates and approval volume.

Data & Quality Insights

307,511
past loan applications
used to train the model

122
pieces of information
captured per application

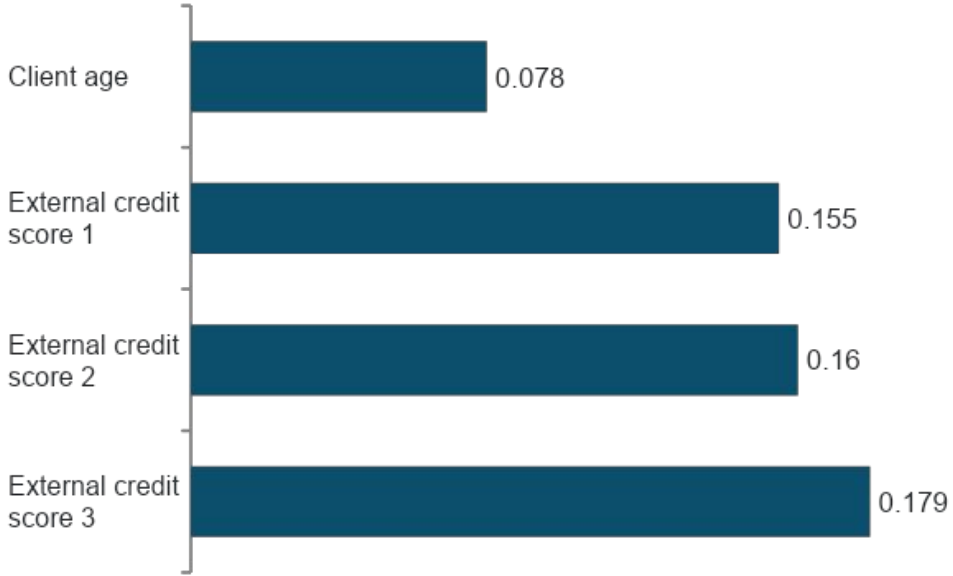
8.07%
of past clients had
payment difficulty



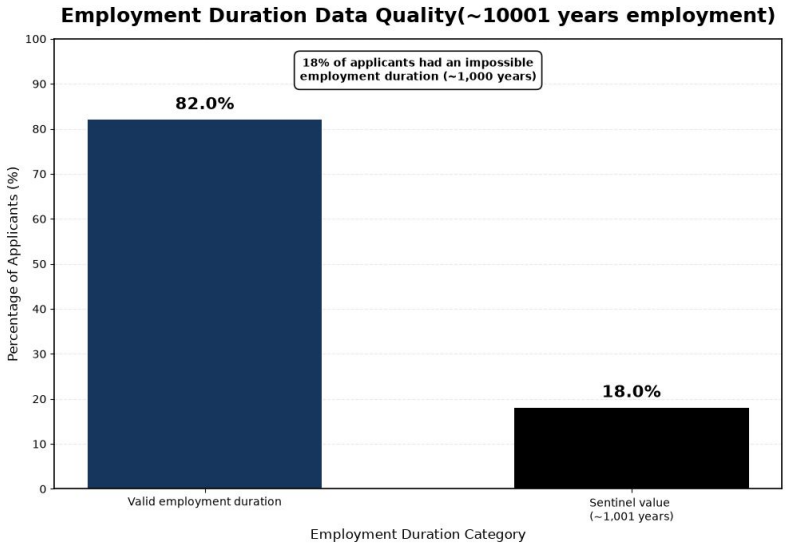
Only ~1 in 12 past applicants defaulted — a strongly imbalanced problem. This matters a lot for how we judge the model.

Early Findings - EDA

Strongest signals of repayment risk

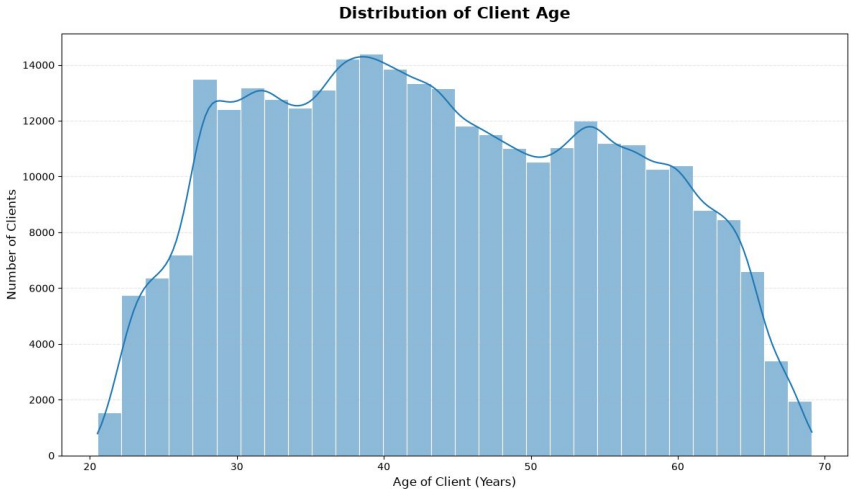


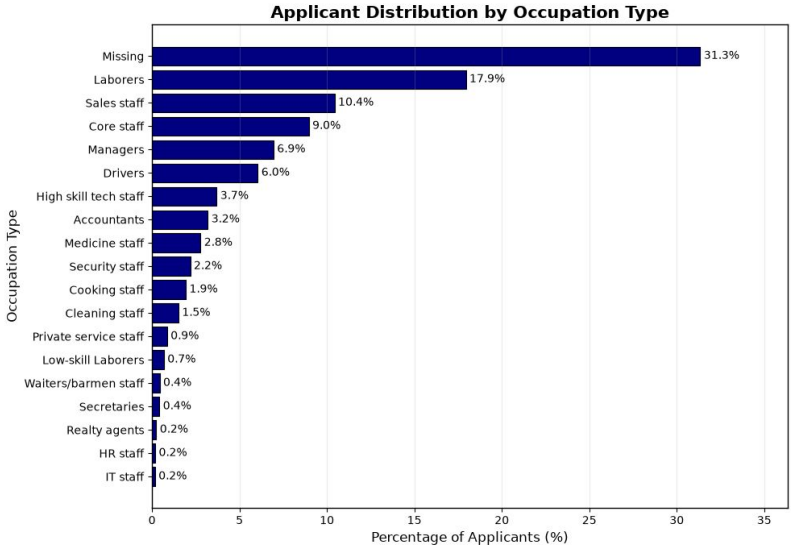
- *Credit bureau score: Higher score → lower default risk.*
- *Age: Younger clients tend to have slightly higher risk.*
- *Older clients: Tend to have slightly lower default risk.*



- *Sentinel values are replaced with NaN before modeling.*

Early Findings - EDA





- *People in the age range 30-40 years are the most applicants which seems pretty normal.*

- *We found 17.9% of labours are defaulters.*
- *Occupation type - found 31.3% missing values.*

Machine Learning Solution

Although the baseline is 91.9% accurate, it catches 0% of defaulters, so accuracy alone is misleading.

The data is highly imbalanced:

- Most applicants do not default.
- A model could get high accuracy by predicting “No Default” for almost everyone.

What matters for the business:

- Recall: Catch as many potential defaulters as possible.
- ROC-AUC / PR-AUC: Measure how well the model separates risky and safe applicants.
- Precision: Avoid flagging too many reliable customers as risky.

Key takeaway:

For loan default prediction, detecting potential defaulters is more important than achieving high accuracy alone.

Model Comparison

We compare every model against the simplest possible baseline: a rule that predicts every applicant will repay.

Approach	Accuracy	Recall on defaulters	ROC-AUC	Precision
Baseline (predict “always repays”)	*0.91*	*0*	—	*0*
Logistic Regression + Balanced Class + Tuning	*0.69*	*0.68*	*0.75*	*0.16*
Decision Tree + Balanced Class +Tuning	*0.66*	*0.67*	*0.72*	*0.15*
Random Forest + Balanced Class + Tuning	*0.70*	*0.62*	*0.72*	*0.16*
XGBoost	*0.69*	*0.69*	*0.76*	*0.17*

Data integrity: *every feature used here is captured strictly before the loan decision — none of it depends on what happens after approval.*

Why We Chose XGBoost?

- **Good risk separation**: ROC-AUC 76% – the model can distinguish higher- and lower-risk applicants fairly well.
- **Finds most risky applicants:** Recall 68.7% – catches about 69% of actual defaulters.
- **Focuses on finding risk:** F2 42% – gives more importance to finding defaulters.
- **Risk Screening:** Of the customers the model **flagged as high risk, 16.6% actually defaulted**.

XGBoost model is good at finding potential defaulters, but the high-risk group still needs further assessment.

Risk Prediction On New Applications

- **Risk Assessment:**
  The model gives each applicant a default risk score.

- **Action:**
  If the risk score is 42% or higher, the application is flagged for further review.

- **Business benefit:**
  This helps the team focus on higher-risk applicants and make consistent decisions.

Note: F2(0.42) = 42% → this threshold was chosen because we wanted to give more importance to finding potential defaulters (Recall) than to Precision.

Potential Enhancements

What We Found

Models have useful predictive signal:

ROC-AUC of 0.75–0.76 shows the features capture meaningful patterns.

XGBoost performed best:

Highest ROC-AUC (0.76), sightly better than Logistic Regression (0.75).

Similar recall across models:

Models identify a comparable proportion of risky customers.

Precision & F1 are low:

Many customers predicted as risky are actually non-risky, creating false positives.

Application data alone is limited:

Without historical credit and repayment information, performance is constrained.

Where We'd Take This Next

Add Historical Data:

Integrate bureau, previous applications, installments, credit card & POS/CASH data and engineer aggregated customer-level features to significantly enhance model quality.

Threshold Based on Business Cost:

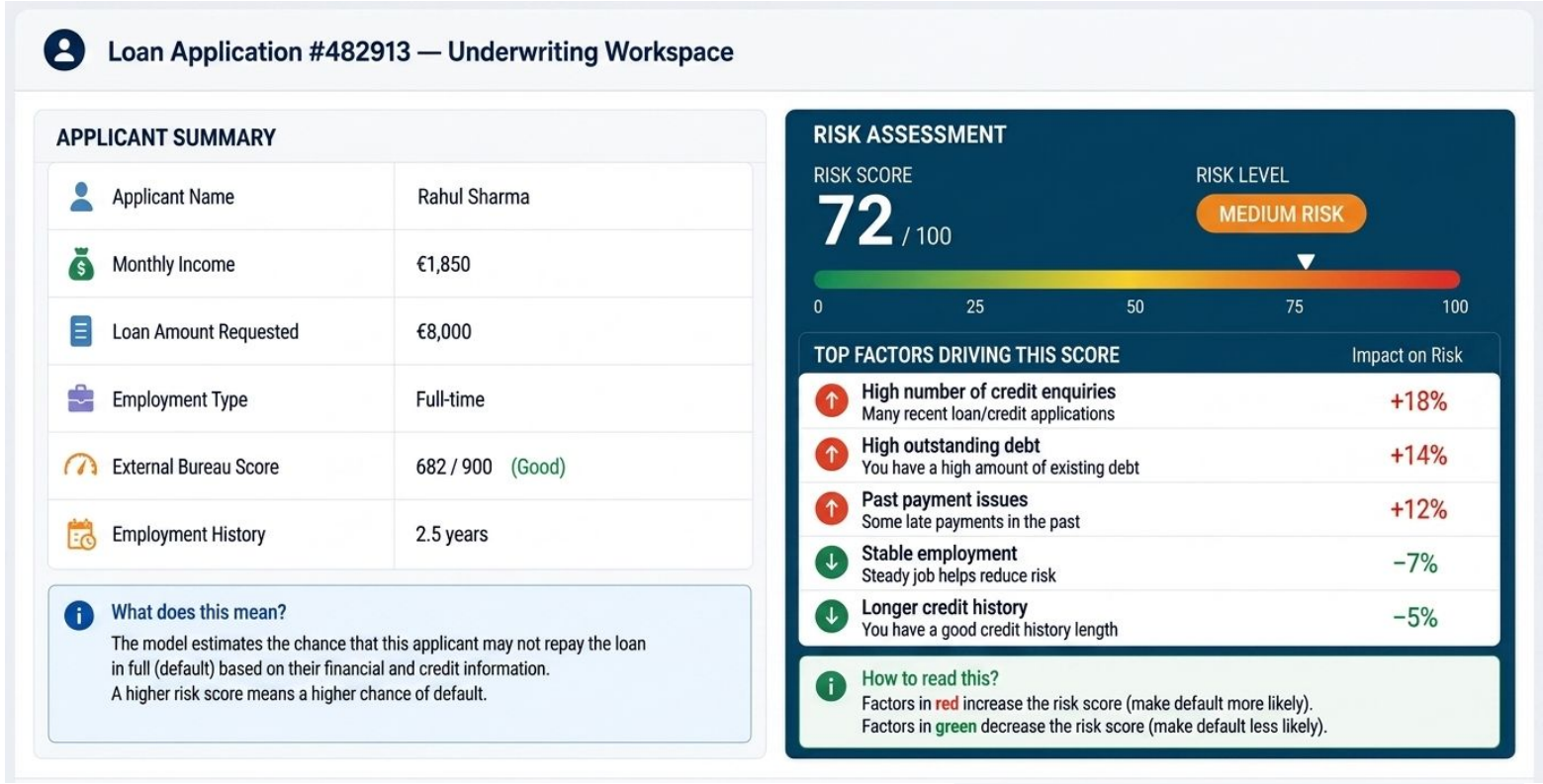
Choose a classification threshold using a cost-sensitive approach that balances false positives and false negatives.

Model Explainability:

Use SHAP / feature importance to explain predictions to build trust and support better credit decisions.

Recommendation : Use the current model as a risk-ranking / decision-support tool rather than an automated loan approval or rejection system.

From Model to Product: Explainable Risk Assessment



Thank you for your time and attention.